impl TopKMeasure for ZeroConcentratedDivergence

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1 Hoare Triple

Precondition

Compiler-verified

- Method `noisy_top_k` *Types consistent with pseudocode.*
- Method `privacy_map` *Types consistent with pseudocode.*

Caller-verified

- Method `noisy_top_k`
  - `x` elements are non-null.
  - `scale` is finite and non-negative.
- Method `privacy_map`
  - `d_in` is non-null and positive.
  - `scale` is non-null and positive.

Pseudocode

```python
# ZeroConcentratedDivergence
def noisy_top_k(x: list[TIA], scale: f64, k: usize, negate: bool) -> list[usize]:
    return gumbel_top_k(x, scale, k, negate)

def privacy_map(d_in: f64, scale: f64) -> f64:
    return d_in.inf_div(scale).inf_powi(ibig(2)).inf_div(8.0)
```

Postcondition

Theorem 1.1. The implementation is consistent with all associated items in the `TopKMeasure` trait.

1. Method `noisy_top_k`:
   - Returns the index of the top element z_i, where each $z_i \sim \text{DISTRIBUTION}(\text{shift} = y_i, \text{scale} = \texttt{scale})$, and each $y_i = -x_i$ if `negate`, else $y_i = x_i$, k times with removal.
   - Errors are data-independent, except for exhaustion of entropy.
2. Method `privacy_map`: For any x, x' where $d_{\text{in}} \geq d_{\text{Range}}(x, x')$, return $d_{\text{out}} \geq D_{\text{self}}(f(x), f(x'))$, where $f(x) = \text{noisy_top_k}(x = x, k = 1, \text{scale} = \text{scale})$.

Definition 1.2. A random variable follows the Gumbel distribution if it has density

$$f(x) = \frac{1}{\beta} e^{-e^{-z}-z} \tag{1}$$

where $z = \frac{x-\mu}{\beta}$, μ is the shift (location) parameter and β is the scale parameter.

Proof of postcondition: ***`noisy_top_k`***. The preconditions of `gumbel_noisy_max` are met, therefore by the postcondition of `gumbel_top_k`, the postcondition of `noisy_top_k` is satisfied. □

Proof of postcondition: ***`privacy_map`***. By Lemma 4.2 of [2], $\mathcal{M}^k_{\text{Gumbel}}(x)$ is equal in distribution to the peeling exponential mechanism, which is the k-fold composition of the exponential mechanism. Proposition 2 of [1] shows that the exponential mechanism satisfies the zCDP privacy guarantee. □

References

[1] Jinshuo Dong, David Durfee, and Ryan Rogers. Optimal differential privacy composition for exponential mechanisms and the cost of adaptivity. *CoRR*, abs/1909.13830, 2019.

[2] David Durfee and Ryan Rogers. Practical differentially private top-k selection with pay-what-you-get composition. *CoRR*, abs/1905.04273, 2019.